Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/05/2019
PI Name:	Boerma, Marjan Ph.D.		
Project Title:	Effects of Chronic and Single-Dose Charged Particle Exposure on Cardiac Function and Structure in Multiple Animal Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	08/07/2017	**End Date:**	01/06/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/6/2020 per NSSC info (Ed., 11/12/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weil, Michael Ph.D. (Colorado State University)		
Grant/Contract No.:	80NSSC17K0425		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent evidence suggests that exposure to high-linear energy transfer (LET) radiation encountered during deep space travel may be associated with short- or long-term risks of cardiovascular disease. However, these risks are not well defined. Previous studies with single doses of high-LET radiation in mouse models have shown small but significant changes in cardiac function and structure. While these studies have started to characterize the cardiovascular effects of low-dose high-LET radiation, due to practical considerations the radiation in these experiments had to be administered in a very high dose rate compared to the chronic low-dose rate radiation exposures that astronauts will encounter in deep space. Therefore, additional work needs to be done in animals that are chronically exposed to low-dose rate high-LET radiation. Moreover, the inclusion of additional animal species will greatly enhance the potential for translation of the results to humans compared to research in mouse models alone. Here, we aim to provide insight into the effects of chronic low-dose rate high-LET radiation exposures on the heart and identify how animal species differ in their response to high-LET radiation. We pursue the following aims: 1) In collaboration with the NASA Specialized Center of Research (NSCOR) on Carcinogenesis, we make use of their long-term studies in male and female mouse models exposed to chronic low-dose rate neutrons to assess cardiac function with echocardiography, and we will collect cardiac tissues samples from these mice to determine connective tissue content, infiltration of inflammatory cells, and mitochondrial function; 2) We perform echocardiography and tissue analysis in male rats that are exposed to low-dose rate neutrons in the neutron facility that is part of the NSCOR; 3) We perform echocardiography and tissue analysis in mice exposed to neutrons and silicon ions and treated with aspirin in their drinking water, to determine whether aspirin can protect against radiation-induced changes in the heart; and 4) We assess protein markers of inflammatory cells, of mitochondria and connective tissue structure in hearts of male rats and rabbits previously exposed to protons and oxygen ions and available in our laboratory. These projects make use of existing animals and tissue samples from NASA-funded laboratories to aid in defining the risk of cardiovascular disease from space radiation exposure, and to determine whether aspirin is a potential countermeasure against cardiovascular disease risk from space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is concern about the potential adverse cardiovascular effects of chronic exposures to low-dose ionizing radiation, such as from medical treatments, occupational low-dose exposures, and radiological accidents. The current project is one of the first to measure cardiac function and structure in animal models of chronic low-dose rate exposures to ionizing radiation. This information will inform not only the risk of radiation exposure during deep-space travel, but also the risks of cardiovascular disease from exposure to low-dose rate radiation exposures on Earth.
Task Progress:	This is the progress report of year 2 of this project. All radiation exposures of mice, rats, and rabbits have been completed, and cardiac function and tissue analysis is underway. Echocardiography was performed on female BALB/C mice ~400 days after initiating chronic exposures to neutrons (to a total dose of 0.12, 0.2, or 0.4 Gy). These studies reveal a significant decrease in left ventricular fractional shortening (indicative of contractile function) and increase in left ventricular posterior wall thickness in animals that received the high (0.4 Gy) dose radiation. These results are suggestive of pathologic overload and hypertrophy. These mice will be reassessed in September 2019 (400 days after completion of the chronic neutron exposures) to determine if the cardiac phenotype evolves with time post-radiation. Animals will be sacrificed soon thereafter for studies on cardiac morphology and mitochondrial function (oxidative phosphorylation (OXPHOS) and release of reactive oxygen species (ROS)). Echocardiography was performed on male C3H mice ~400 days after initiating chronic exposures to neutrons (to a total dose of 0.12, 0.2, or 0.4 Gy). Results show a clear impairment in left ventricular fractional shortening and increase in left ventricular wall thickness, consistent with the neutron-irradiated female BALB/C mice described above. These mice will undergo another cardiac ultrasound in November 2019 (400 days after completion of the chronic neutron exposures) to determine if the cardiac phenotype evolves with time post-radiation. The mice will be sacrificed soon thereafter for studies on cardiac morphology and mitochondrial function (OXPHOS and ROS release). Male Long-Evans rats were echoed in March 2019, immediately after 400 days of chronic neutron exposure (total dose of 0.4 Gy). These studies showed no significant effect of irradiation on left ventricular function or morphology. However, a small, but statistically significant increase in isovolumic relaxation time (indicative of slower myocardial relaxation) was seen – perhaps representing early stages of left ventricular stiffening and diastolic dysfunction (which typically precede wall thickening and systolic dysfunction). These rats will be reassessed in early January 2020 (400 days after completion of the chronic neutron exposures) to determine if the cardiac phenotype evolves with time post-radiation. Rats will be sacrificed soon thereafter for studies on cardiac morphology and mitochondrial function (OXPHOS and ROS release). At day 400 after a single dose of silicon ions of 0.4 Gy, echocardiography was performed on male C3H mice that received regular drinking water, or drinking water containing aspirin. While we detected no differences in left ventricular volumes or contractile function, we did observe a non-significant trend of higher left ventricular posterior wall thickness in the irradiated group that was attenuated by aspirin treatment. The inhibition by aspirin suggests that the trend of a changed posterior wall thickness is due to an inflammatory mechanism. These mice will soon be reassessed for cardiac structure/function at day 800 (June 2019) to determine if these trends worsen or manifest differently as animals age. We will also assess cardiac mitochondrial function and ROS release at this time (terminal experiments; late June 2019), and can follow up with histological and biochemical analyses of tissue remodeling and inflammation. Male New Zealand White rabbits were exposed to 0.5 Gy of protons (250 MeV) or 0.5 Gy of oxygen ions (600 MeV/n), or sham-irradiation, and echocardiography was performed at 10 and 12 months after irradiation. In year 2 of this project, we have completed the analysis of the echocardiography recordings. Left ventricular interior diameter in systole and diastole and the related parameters end-diastolic and end-systolic volume were significantly reduced at 10 months after protons and oxygen ions compared to age-match sham-irradiated controls. At 12 months, only left ventricular inner diameter in systole and end-systolic volume in the oxygen ion group remained significantly reduced. Ejection Fraction and Fractional Shortening were significantly increased in both radiation groups at 10 months, and the same trend was seen for the oxygen ion group at 12 months. Tissue specimens of rabbit heart collected at 12 months after irradiation were subjected to histological staining. Histopathological analysis revealed no indications of acute inflammation, acute myofiber necrosis, microvascular damage, chronic interstitial fibrosis, chronic epicardial fibrosis, or coronary artery and valvular damage. Moreover, there was no change in total collagen deposition in the heart or aorta.

	Tissue specimens of rabbit heart collected at 12 months after irradiation were also analyzed by Western-Blotting to assess the expression of thrombin, a protease known to be upregulated chronically after high-dose irradiation and thought to play a role in radiation-induced endothelial dysfunction. At 12 months after exposure to protons, a significant increase in left ventricular thrombin expression was seen compared to age-matched sham-irradiated controls. Oxygen ions did not induce left ventricular thrombin levels.
Bibliography Type:	Description: (Last Updated: 01/08/2026)
Abstracts for Journals and Proceedings	Boerma M, Cao M, Sridharan V, Singh P, Landes RD, Chicco AJ, Weil MM. "Effects of chronic and single-dose charged particle exposure on cardiac function and structure in multiple animal models." Space radiation: cardiovascular disease and late CNS effects. Presented at the NASA Human Research Program Workshop, Galveston TX, January 22-25, 2019. Abstracts. NASA Human Research Program Workshop, Galveston TX, January 22-25, 2019 , Jan-2019
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Articles in Peer-reviewed Journals	Seawright JW, Sridharan V, Landes RD, Cao M, Singh P, Koturbash I, Mao XW, Miousse IR, Singh SP, Nelson GA, Hauer-Jensen M, Boerma M. "Effects of low-dose oxygen ions and protons on cardiac function and structure in male C57BL/6J mice." Life Sciences in Space Research. 2019 Feb;20:72-84. https://doi.org/10.1016/j.lssr.2019.01.003 ; PubMed PMID: 30797436; PubMed Central PMCID: PMC6391741 , Feb-2019
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